AI vs Machine Learning

AI:

AI is mimicking human brain, it is the wide field.
Examples:

- Chatbots
- Face recognition

AI Includes:

- Machine Learning
- Deep Learning
- Natural Language Processing (NLP)
- Computer Vision

ML:

ML is a subset of AI, learn patterns from data.
ML Workflow:

1. Collect data.
2. Clean and preprocess data.
3. Train a model.
4. Evaluate performance.
5. Make predictions on new data.

Machine Learning vs Deep Learning

Deep Learning

- A subset of Machine Learning.
- Uses Artificial Neural Networks with many layers.
- Requires large datasets.
- Needs high computational power (GPUs).

Supervised vs Unsupervised vs Reinforcement Learning

Supervised Learning

The model learns from **labeled data**, where the correct answer (target) is already known.
Applications:

- Email spam classification
- House price prediction

Tasks:

- Classification for categorical data
- Regression for numerical data

Unsupervised Learning

The data has **no labels**.
The algorithm tries to discover hidden structures or patterns.

Tasks:

- Clustering : Groups similar data together.
- Dimensionality Reduction : Reduces the number of features while preserving important information.

Reinforcement Learning

An agent learns by interacting with an environment.
Instead of labels, it receives rewards or penalties
The goal is to maximize total reward.